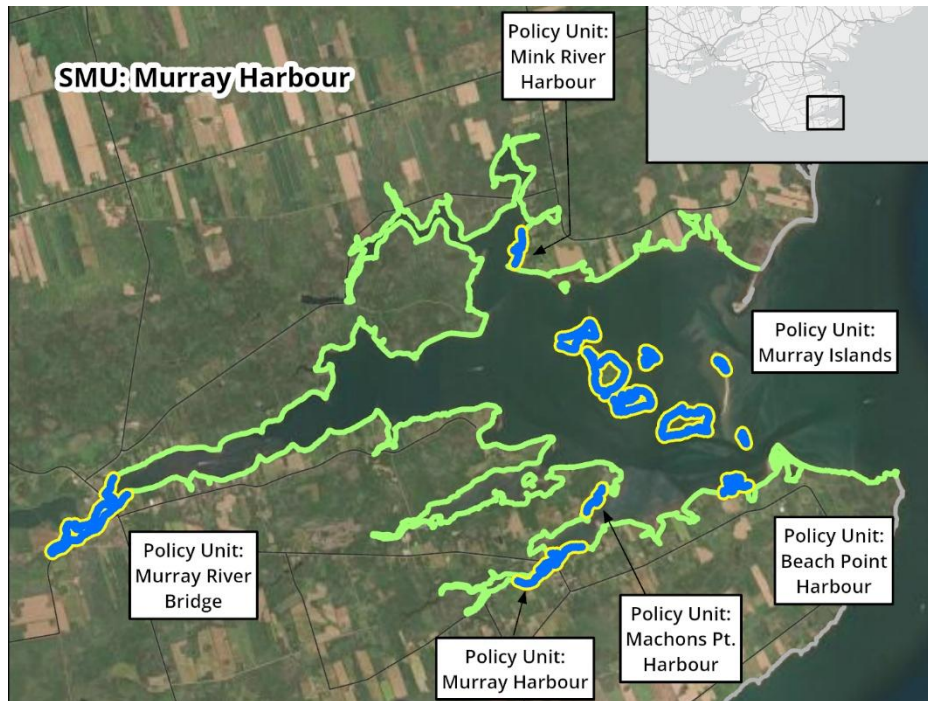


3. Murray Harbour

Shoreline length: 92 km



Shore Types

Murray Harbour contains multiple estuaries including one connected to the Murray River. Located on the east end of PEI, with a majority of low plains and wetlands.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
61	5	4	2	23	2	2	1

Governance, Infrastructure, and Economy

The Murray Harbour SMU includes the rural municipalities of Murray Harbour and Murray River (municipalities with no planning, ~15% combined), with the remaining ~85% being unincorporated.

Coastal infrastructure includes active wharves and the roads leading to them, including Point Pleasant Road (Route 17); Girl Guide Camp Road (Murray River); the Point Pleasant Road Bridge (Little Basin); Route 17 (Cambridge); Mink River Road (Murray Harbour); and Beach and Poverty Roads (Murray Harbour North).

Other infrastructure includes the Beach Point Lighthouse; Wharf Lane, Beach Point, the Murray Harbour Range Rear Light; Keeping Road (Murray Harbour, at the beach); Murray Lane (Murray Harbour); Main Street and Riverside Drive; Westview Drive; the bridge and Confederation Trail crossing over the Murray River; the Murray River Sewage Lagoon; Machon Point Road; Fox River Road; Gladstone Road (Route 348, Murray Harbour); Route 4 Main Street (Murray River); the Murray River dam; Marina, Wharf at Big Basin, Murray Harbour North and Livingstone Road (Route 348, Murray River).

Coastal economic drivers include the Seal Cove Campground (Murray Harbour North), the Murray Harbour North wharf, Machon Point Road wharf and industry, the Murray Harbour wharf and marina, and the Beach Point Lobster Processing Company and wharf (Murray Harbour). Local and legacy land use constraints include several undeveloped small-lot subdivisions, with several lots (many undeveloped), and unpaved roads.



Cliffs updrift of Beach Point



Beach Point Harbour



Machons Point Harbour



Murray River Bridge

Natural Assets and Heritage

Coastal parks and cultural sites include Kings Castle Provincial Park, Murray River Memorial Park (playground), Railhead Park (playground), the Beach Point Lighthouse, and Black Brook Lookout Point (hiking/park).

The shoreline includes critical habitat for bird species including the endangered Piping Plover.

Murray Islands is a cluster of undeveloped islands including, Cherry Island and Gordons Islands, within Murray Harbour, of total shoreline length 12 km. The islands are owned by the Government of Prince Edward Island and the Nature Conservancy of Canada. The NCC ownership indicates a conservation designation, though no named natural area or protected area is identified in available data.

Planning and Policy Framework

Neither Murray Harbour or Murray River have official plans and provincial rules apply throughout this SMU. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a

calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

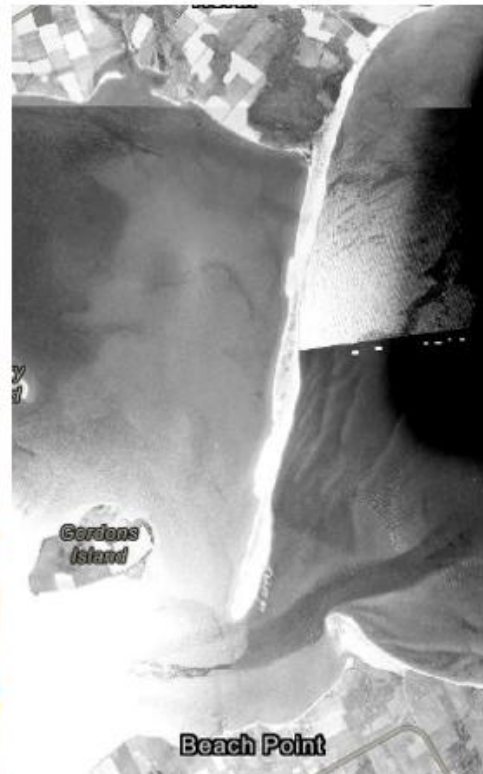
Sediment transport and Morphology

While the estimated net sediment transport rate inside the Bay is low, it is high along the outer islands and exposed portion of Beach Point, which feeds sand formations at the entrance of the bay. Historical air photos show the vulnerability to breaches of barrier islands at the entrance of the bay, which highlights the importance of maintaining sediment supply from Murray Head to the South and Irwings Cape to the North (situated in adjacent Pembroke Grahams Pond SMU)

1765



1935



1968



2000



Coastal Flood and Erosion Risk

Infrastructure along low plain, wetlands, and sand dune shorelines are vulnerable to increasing flood hazards. Erosion hazard is also highest along sand dunes and wetlands. Wetland migration should be allowed to follow rising sea levels to avoid coastal squeeze

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline].

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	61 [1/km]	74 [1/km]	88 [1/km]	111 [1/km]	2 [<1/km]	12 [<1/km]	38 [<1/km]
Roads	4121 m [40 m/km]	5611 m [54 m/km]	6871 m [66 m/km]	10460 m [100 m/km]	375 m [4 m/km]	760 m [7 m/km]	1779 m [17 m/km]

Proposed SMP Strategies

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
3.00 Murray Harbour	NAI	MR	MR
Rationale	The proposed short-term strategy for Murray Harbour is No Active Intervention. This is particularly important for outer areas supplying sediment to the beaches fronting the bay, such as Beach Point which protects the harbour. In areas with at-risk infrastructure, Managed Realignment is advised in later epochs.		
3.01 Beach Point Harbour -PU	R	R	R
Rationale	Resist strategy would be suitable for key harbour infrastructure. Future works will also require flood mitigation, subject to final strategies to be developed with the local authority.		

3.02	Machons Pt Harbour -PU	R	R	R
	Rationale	Resist strategy would be suitable for key harbour infrastructure. Future works will also require flood mitigation, subject to final strategies to be developed with the local authority.		
3.03	Mink River harbour -PU	R	R	R
	Rationale	Resist strategy would be suitable for key harbour infrastructure, but will not mitigate long-term flood risk, including flood isolation during extreme storms. Future works would also require flood mitigation, subject to final strategies to be developed with the local authority.		
3.04	Murray Harbour -PU	R	R	R
	Rationale	Resist strategy would be suitable for key harbour infrastructure, but will not mitigate long-term flood risk, including flood isolation during extreme storms. Flood mitigation (raising assets or relocation) is most notably required for the Murray Harbour Fire Hall on the south side of the bridge.		
3.05	Murray Islands -PU	NAI	NAI	NAI
	Rationale	A No Active Intervention strategy is recommended for all three epochs. This will help preserve natural habitat and shoreline processes, including the sheltering function for Murray Harbour. The local shoreline and hazard zone should remain natural, undeveloped.		
3.06	Murray River bridge -PU	R	R	R
		Resist strategy would be suitable for key infrastructure, but will not mitigate long-term flood risk. Future works would also require flood mitigation along Main Street north of the bridge.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.